Task 3: Research Problem Formulation & Idea Generation

Base Paper: SED Paper| Group 6

Methodology

Three frameworks inform how problems and ideas are generated below:

- **HypER** (Vasu et al.) : a hypothesis is only as good as the reasoning chain behind it. Every idea below is traced to a specific limitation (from the prior assignment) and the literature that grounds it, no idea is asserted without a chain: *limitation → why unsolved → proposed mechanism*.
- **ResearchAgent** (Baek et al.) : decomposes idea generation into **problem → method → experiment design**, and scores each idea on five human-aligned criteria. We adopt this structure and scoring axes (Clarity, Originality/Novelty, Feasibility, Significance/Impact) for ranking.
- **HypoEvolve** (Liu et al.) : treats hypotheses as a *population* refined by specialized critique roles (mechanistic argument, assumption-check, evidence/testability). We use this lens when ranking: each idea is stress-tested for whether its mechanism is sound and whether it's testable within a course-scale project, not just whether it "sounds novel."

Source gaps used (from the prior limitations assignment (Link) all grounded in SED's paper text or citing works already surveyed: DPSeg, SCAN, S-Seg, ESC-Net):

- L3: Near-synonym category confusion (author-stated in SED)
- L4: Category Early Rejection (CER) uses a fixed, non-adaptive top-k

These two are chosen for the research problems below because they are **not yet addressed by any paper in our literature survey**, DPSeg attacks L1/L2 (modality gap, upsampling), SCAN attacks L5 (domain bias), ESC-Net attacks L7 (boundary quality). L3 and L4 remain open.

Sub-task 1: Research Problem Statements

Research Problem 1 (RP1)

How can SED's cost-map construction be modified to disambiguate near-synonym categories (e.g., water/sea, earth/rock, car/truck) at inference time, without retraining CLIP or degrading SED's open-vocabulary generalization?

It names the exact failure (near-synonym pairs), the exact component responsible (cost-map construction), and the exact constraint any solution must respect (no CLIP retraining, no generalization loss),]. This is testable directly on SED's existing checkpoint and ADE20K/PASCAL-Context confusion pairs.

Grounding: SED's own Future Work section admits this failure exists but proposes no mechanism. No paper in our survey (DPSeg, SCAN, S-Seg, ESC-Net, PnP-OVSS, LPOSS) targets category-level semantic disambiguation — they target spatial/modality-gap problems instead. This is a genuine open gap, not a rehash of surveyed work.

Research Problem 2 (RP2)

How can Category Early Rejection (CER) select a scene-adaptive top-k threshold instead of a fixed k=8, so that cluttered scenes are not under-pruned and simple scenes are not over-computed, while preserving SED's reported inference speedup?

It targets one specific hyperparameter (k in CER) in one specific module (SED's decoder), with a measurable trade-off (accuracy on complex scenes vs. speed on simple scenes) that can be benchmarked directly against SED's own reported numbers (4.7× speedup, mIoU on A-847/A-150).

Grounding: SED reports k=8 as a single global setting chosen empirically; no ablation or mechanism for per-scene adaptation exists in the paper or in any surveyed follow-up work. This is an algorithmic gap independent of the cost-map-quality problems DPSeg/SCAN address.

Sub-task 2: Top-5 Ranked Research Ideas per Problem

Ranking criteria (adapted from ResearchAgent's human-aligned rubric, 1-5 scale each):

- **Novelty** : how distinct from SED and all surveyed follow-ups
- **Feasibility** : implementable and testable within a course-scale project (no full CLIP retraining, no massive new datasets)
- **Clarity** : how precisely the mechanism is specified
- **Impact** : how much it would matter if it worked

RP1: Near-Synonym Disambiguation: 5 Ranked Ideas

1) **Idea - Inter-class margin loss on confusable category clusters**

   a) **Method:** Compute pairwise cosine similarity between all COCO-Stuff (171) category text embeddings. Cluster pairs above a similarity threshold (e.g., >0.9) as "confusable sets" (water/sea, earth/rock, etc. This reproduces SED's own stated failure cases). During SED's existing fine-tuning, add an auxiliary margin loss term: for images containing ground-truth pixels from two categories in the same confusable cluster, penalize the model if their cost-map activations at the corresponding pixels are closer than a margin δ. This only touches training loss and no architecture change.

   b) Codifiable next step: Implement as an additional loss term added to SED's existing pixel-wise cross-entropy; requires only the confusion-cluster list (computable once) and existing COCO-Stuff training pipeline.

   c) Scores: Novelty 3 | Feasibility 5 | Clarity 5 | Impact 4

2) **Idea - Confusable-pair contrastive projection head Method:**

   a) **Method:** Freeze CLIP entirely. Add a small trainable linear projection layer applied only to text embeddings of categories flagged as confusable. Train this projection with a triplet loss (anchor = category name, positive = correct pixel region embedding, negative = confusable category's pixel regions) using SED's existing training images. At inference, only confusable-category embeddings pass through the projection; all others use raw CLIP embeddings — preserving open-vocabulary generality for the 99% of categories with no confusion issue.
   b) **Codifiable next step:** A single linear/MLP layer (few hundred parameters), trained separately from SED's main decoder, activated conditionally at inference.
   c) **Scores:** Novelty 4 | Feasibility 4 | Clarity 4 | Impact 4

3) **Idea - Targeted visual exemplar retrieval for confusable classes only**

   a) **Method:** Rather than full dual-modal cost volumes (DPSeg's approach, applied to *every* category), apply visual prompting selectively: for the small set of categories flagged as confusable (from SED's own confusion matrix), retrieve 3–5 visual exemplar image crops per category (from COCO-Stuff training images) and compute an auxiliary image-image cost map only for these categories, fused with the existing text-based cost map via a learned gate. This keeps SED's efficiency intact for the ~95% of non-confusable categories.
   b) **Codifiable next step:** Requires building a small exemplar bank (one-time offline step) and a lightweight fusion gate (1 conv layer), most implementation-heavy of the five but well-specified.
   c) **Scores:** Novelty 4 | Feasibility 3 | Clarity 4 | Impact 5

4) **Idea - Hypernym-augmented compositional prompts**
   a) **Method:** For each confusable category, auto-generate 2–3 augmented text prompts using WordNet hypernym/definition data (e.g., "water" → "a body of liquid, not solid ground"; "sea" → "a large body of saltwater connected to an ocean"). Compute cost maps for each augmented prompt variant and average them per category before decoding. This adds discriminative context directly in the text signal SED already consumes. No architecture change, only prompt engineering + multiple forward passes for confusable classes.
   b) **Codifiable next step:** WordNet lookup (via NLTK) is off-the-shelf; only the prompt-averaging logic needs to be added to the existing text-encoding pipeline.
   c) **Scores:** Novelty 3 | Feasibility 5 | Clarity 4 | Impact 3

5) **Idea - Post-hoc confusion-aware re-ranking classifier**
   a) Method: Train a small secondary classifier (e.g., a 2-layer MLP) on local visual context features (average-pooled encoder features in a small window) that activates only when SED's decoder's top-2 predicted categories are both members of a confusable cluster, re-ranking between just those two. This is a lightweight patch applied post-decoding, touching no part of SED's trained weights.
   b) Codifiable next step: Simplest to implement (single small classifier, trained on SED's own validation set predictions), but the least architecturally interesting.
   c) Scores: Novelty 2 | Feasibility 5 | Clarity 5 | Impact 3

RP2: Scene-Adaptive Category Early Rejection: 5 Ranked Ideas

1) **Idea - Learned k-predictor head**
   a) **Method:** Add a lightweight auxiliary head (small MLP or 1×1 conv) on top of the coarsest-resolution cost-map features (already computed inside SED's existing pipeline, at 1/32 resolution). This head is trained to directly regress the optimal k for CER at that decoder layer, supervised using labels generated offline: for each training image, sweep k ∈ {4, 8, 16, 32, all} and record which k preserves ground-truth mIoU while minimizing computation — the smallest k that doesn't hurt accuracy becomes the training label.
   b) **Codifiable next step:** Requires one offline sweep over the training set to generate k-labels (straightforward script), then training a small auxiliary head jointly or after SED's main training, clean, testable, directly measurable against SED's reported 4.7× speedup.
   c) **Scores:** Novelty 4 | Feasibility 4 | Clarity 5 | Impact 5

2) **Idea - Confidence-entropy heuristic k (no training required)**
   a) **Method:** At the coarsest decoder layer, compute the entropy of the per-pixel class-score distribution from the initial cost map. Aggregate into a single scene-complexity score (e.g., count of categories with avg confidence above a fixed threshold across the image). Set k dynamically per image using a simple rule: k = clip(count + margin, k_min=4, k_max=32). No training needed. It is purely a runtime heuristic computed from existing intermediate activations.
   b) C**odifiable next step:** Fastest to implement, a few lines of code inserted into SED's existing decoder forward pass, no new training data or labels required.
   c) **Scores:** Novelty 3 | Feasibility 5 | Clarity 5 | Impact 3\

3) **Idea - Two-pass adaptive CER**
   a) **Method:** Run a fast first pass with a large fixed k (e.g., k=32) purely to estimate the number of categories actually present in the scene (via a coarse histogram of high-confidence categories). Use this count to set k for a second, full-resolution decoder pass. Simple scenes get a small k (fast); complex scenes get a larger k (accurate), but every image now pays the cost of the cheap first pass.
   b) **Codifiable next step:** Requires running the decoder's early layer twice per image; well-specified and simple to benchmark against SED's single-pass baseline for a clear speed/accuracy trade-off comparison.
   c) **Scores:** Novelty 3 | Feasibility 5 | Clarity 4 | Impact 3

4) **Idea - Cluster-level rejection (shared mechanism with RP1)**
   a) **Method:** Instead of rejecting individual categories at the CER stage, first group categories into semantic clusters (the same confusable-cluster computation used in RP1 Idea 1, reusable across both problems). Apply CER at the cluster level: reject entire clusters early, but retain all members of surviving clusters through to the final decoder layer. This scene-adapts naturally cluttered scenes with many distinct clusters retain more categories; simple scenes with few clusters prune aggressively.
   b) **Codifiable next step:** Reuses RP1's cluster computation; requires modifying CER's rejection logic to operate on cluster IDs rather than raw category indices; moderate implementation effort, clean synergy with RP1.
   c) **Scores:** Novelty 4 | Feasibility 3 | Clarity 3 | Impact 4

5) **Idea - RL policy for per-layer, per-image k selection**
   a) **Method:** Train a reinforcement learning policy (e.g., a small policy network) that selects k independently at each decoder layer for each image, optimizing a reward combining mIoU and latency. The policy observes the same coarse cost-map features as Idea 1 but is optimized end-to-end via policy gradient rather than direct regression supervision, allowing it to discover non-obvious per-layer k schedules.
   b) **Codifiable next step:** Most complex to implement. It requires defining a reward function, a training loop with policy gradient updates, and careful tuning to avoid instability; feasible as a stretch goal but risky for a course-scale timeline.
   c) **Scores:** Novelty 5 | Feasibility 2 | Clarity 3 | Impact 4

Summary Table

Problem	Idea	Novelty	Feasibility	Clarity	Impact
RP1	Margin loss on confusable clusters	3	5	5	4
RP1	Contrastive projection head	4	4	4	4
RP1	Targeted visual exemplar retrieval	4	3	4	5
RP1	Hypernym-augmented prompts	3	5	4	3
RP1	Post-hoc re-ranking classifier	2	5	5	3
RP2	Learned k-predictor head	4	4	5	5
RP2	Confidence-entropy heuristic k	3	5	5	3
RP2	Two-pass adaptive CER	3	5	4	3
RP2	Cluster-level rejection	4	3	3	4
RP2	RL policy for k selection	5	2	3	4

Grounded in: SED (Xie et al., CVPR 2024) and prior limitations analysis (L3, L4). Methodology adapted from HypER (Vasu et al., 2025), ResearchAgent (Baek et al., 2024), and HypoEvolve (Liu et al., 2026).